



OPEN Hemocompatibility of the Realheart TAH against SynCardia TAH in full-body circulation with human blood

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The SynCardia Total Artificial Heart (TAH) is FDA-approved for bridge to transplantation and has been implanted in > 2,100 patients. Clinical data demonstrate hemolysis whilst preserving von Willebrand factor (VWF), but no published in vitro blood damage profile exists. Such baseline data are important for developing and evaluating next-generation TAH systems. The Realheart TAH is a four-chamber pulsatile positive displacement TAH which mimics the natural heart's atrioventricular plane movement for physiological flow patterns and gentle blood handling. This study compares the hemocompatibility of SynCardia and Realheart in a full-body circulation test setup. Hemolysis testing of pooled human blood was performed according to ASTM standards in a mock loop simulating systemic and pulmonary circulation. Complete blood counts were obtained using automated hematology analysis. Plasma samples were assessed for VWF activity and antigen by ELISA. Mean mgNIH ($n=6$) was 226.0 ± 30.6 mg/100 L for SynCardia and 35.5 ± 4.3 mg/100 L for Realheart ($p=0.002$), a 6.36-fold increase in SynCardia. MIH values were 20.80 ± 2.65 and 3.27 ± 0.37 , respectively ($p=0.002$). SynCardia significantly increased cell-derived fragments compared to Realheart, whilst both devices preserved VWF. Realheart demonstrated substantially improved hemocompatibility under matched full-body circulation conditions.

Keywords Realheart, SynCardia, Total Artificial Heart, Hemolysis, Hematology, Von Willebrand Factor

Abbreviations

AVWS	acquired von Willebrand syndrome
BPX-80	continuous-flow left ventricular assist device positive control
CF-LVAD	continuous-flow left ventricular assist device
CVP	central venous pressure
mgNIH	normalized index of hemolysis
MIH	modified index of hemolysis
MP	microparticle
MPV	mean platelet volume
pfHb	plasma free hemoglobin
RDW _a	red cell distribution width, absolute
RH	Realheart
SYN	SynCardia
TAH	total artificial heart
VWF	von Willebrand factor
VWF:Ag	von Willebrand factor antigen
VWF:CB	von Willebrand factor collagen-binding activity

Heart failure remains a global health burden, affecting an estimated 55 million individuals worldwide and associated with substantial morbidity and mortality¹. The prognosis remains poor, with a median 5-year survival

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near 50%². Limited donor availability (~10,000 transplants/year) sustains the need for total artificial heart (TAH) therapy.

SynCardia is the only FDA-approved TAH and has been implanted in >2,100 patients. However, its use is associated with complications including hemolysis, bleeding, infection³, and progressive organ dysfunction^{4,5}. Clinical studies have demonstrated evidence of significant hemolysis in TAH recipients, characterized by low haptoglobin levels, elevated lactate dehydrogenase, and detectable plasma free hemoglobin (pfHb)^{3,6}, indicating ongoing blood trauma³.

In vitro hemocompatibility testing according to ASTM standards enables controlled evaluation of blood trauma. Recent mock circulation systems enable testing of pulsatile devices under physiologic loading conditions. However, most prior studies have focused on systemic (left-sided) circulation^{7,8}, and comparative data evaluating complete TAH systems under combined systemic and pulmonary conditions remain limited.

The Realheart TAH is a fully pulsatile four-chamber device designed to reproduce physiological atrioventricular plane displacement⁹ and generate physiological flow patterns^{10–13}. Previous in vitro testing using human blood under systemic conditions demonstrated lower hemolysis than SynCardia⁸. The present study extends this work by evaluating full-device performance in a modified circulation loop enabling simultaneous systemic and pulmonary circulation, and by assessing von Willebrand factor (VWF) function as an additional marker of hemocompatibility.

Bleeding is a common early complication following TAH implantation⁵. Unlike continuous-flow devices, where acquired von Willebrand syndrome (AVWS) is well described, pulsatile TAHs appear to preserve VWF function in clinical settings^{14–16}, although findings have been variable and in vitro data are limited. Therefore, this study also examines the effects of both TAH systems on VWF activity in a controlled experimental model.

Materials and Methods

Two pulsatile total artificial heart systems, the Realheart TAH prototype V11.4 (RH) and the SynCardia (SYN), were evaluated in a mock circulation loop configured to simulate combined systemic and pulmonary circulation. The setup was adapted from previously described systemic-only models^{7,8} with modifications to enable full-body circulation testing.

Mock circulation loop

Two systemic loops described previously⁸ were modified and combined to create a recirculating systemic and pulmonary loop (Fig. 1A). The modifications to the pulmonary side included adding additional atrial compliance capacity by a second compliance vessel (25 L flask with a modified air-tight cap connected to a manometer) and increasing circuit tubing diameter from 3/8" to 1/2".

Pressure was applied to the mock loop via both aortic and atrial compliance chambers, regulated using aneroid manometer gauges connected to the counter pressure cylinders and second atrial compliance vessel, respectively.

A circulating heated water bath (EcoLine Staredition 003 E 103, LAUDA, Lauda-Königshofen, Germany) connected to the atrial compliance chamber maintained temperature, while the arterial compliance chamber was half-filled with lukewarm water.

Left and right TAH pumps were connected to the polymethyl methacrylate (PMMA) compliance chambers housing 1.5 mm thick Elastosil RT 625 silicone bellows representing left atrium and aorta; right atrium and pulmonary artery, ac.biomed, Aachen, Germany^{7,8}, Fig. 1B and C. SYN connected directly to the chamber ports; RH was attached using custom vacuum-cast Hei-Cast 8434 N connectors (Arrk Taiwan Co., Ltd, New Taipei City, Taiwan).

Preload and afterload were measured by BD Venflon IV PVK 18G 45 mm (Becton, Dickinson and Company, New Jersey, United States) connected to pvb DPT6100 pressure sensors (CODAN pvb critical care GmbH, Forstinning, Germany) in the silicone bellows at the pump interface to ensure a measurement close to the valves.

Polyvinyl chloride (PVC) 1/2" x 3/32" (ECC NODOP Bloodline, Raumedic AG, Helmbrechts, Germany) tubing divided into three standardized sections of lengths (4.5, 5.6 and 11.3 cm) and one device-dependent section to ensure equal test rig blood volume (SYN: 115.6 cm, RH: 16.9 cm) were joined by three 1/2" x 1/2" Luer lock ports (ECC-connector, Raumedic AG). Blood sampling was performed via the 1st port, filling/de-airing via the 2nd port, and temperature monitoring via the 3rd port using a thermocouple probe (RS PRO Type K Thermocouple 3 m Length, 1/03 mm Diameter → +260 °C, Corby, UK). Flow was measured using a BioProTT FlowTrack SL system with a corresponding BioProTT Clamp-On Transducer BCT 1/2 x 3/32" A (em-tec GmbH, Finning, Germany) positioned upstream of a resistance clamp that fine-tuned pressure head (BioValve, Qosina, Ronkonkoma, United States).

Blood preparation

Fresh human whole blood (<48 h post-collection) from healthy donors was obtained from Karolinska University Hospital. Donors provided consent at the time of donation for research use. Blood units were collected and managed by the blood center for clinical purposes, and surplus units were subsequently made available for research. Samples were fully anonymized prior to receipt and pooled before experimentation, precluding identification of individual donors. The use of human material was assessed by the Swedish Ethical Review Authority (Etikprövningsmyndigheten), which determined that ethical approval was not required. All procedures were conducted in accordance with relevant institutional and national guidelines and regulations and conformed to the principles of the Declaration of Helsinki. Blood was prepared and handled according to ASTM F1830 guidelines. Units were pooled by ABO and RhD group, quality-controlled, and adjusted to a hematocrit of 29–32% using a buffered solution to maintain physiological pH.

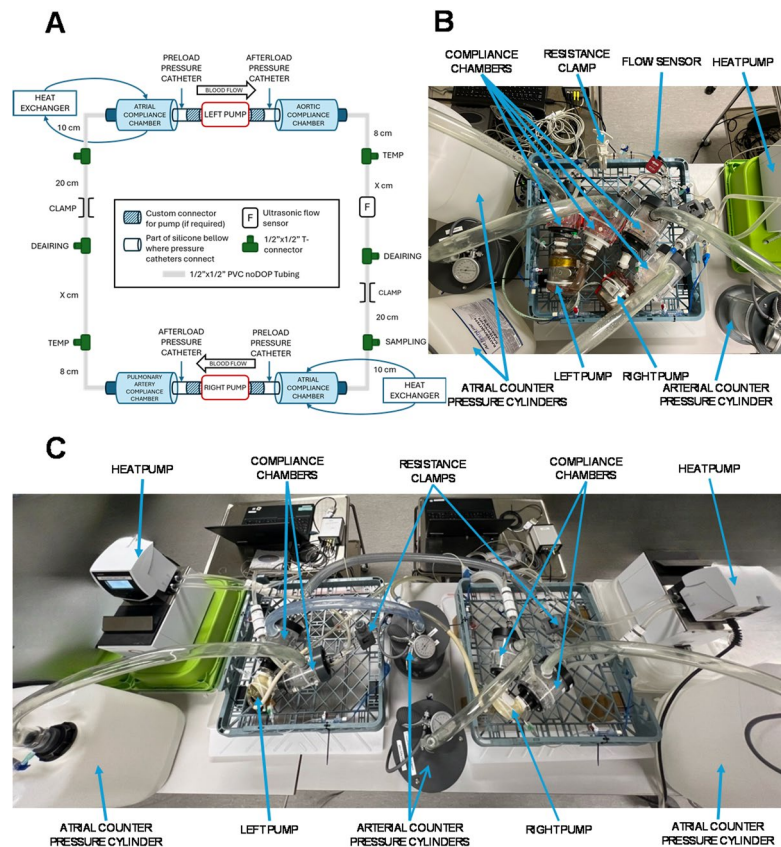


Fig. 1. Mock full-body circulation loop configuration and test rig setups. (A) Both left and right pumps of the TAH are connected to custom-made connectors (for the RH) or directly to the bellows (SYN) which are housed in their respective compliance chambers. The length of section X is customized to the pump used to maintain the same loop blood volume for the test and comparator device. To form a full-body loop, the outflow of the pulmonary circulation loop is connected to the inflow of the systemic circulation loop and vice versa. (B) RH and (C) SYN full-body circulation rig running during a saline test.

Prior to testing, blood was stored under controlled conditions and equilibrated before introduction into the circuit. Each experiment used a shared blood pool for both devices and the corresponding static control.

Test design and sample size

Paired testing using the same blood pool for devices and static control was performed on six occasions ($n=6$). Samples for VWF analysis were obtained from the last 5 experiments ($n=5$).

Test rig preparations

Phosphate-buffered saline (PBS) was pumped through the circuit for 20 min to wet the surfaces, drained, and replaced with 1050 mL of blood. Target temperature, flow and differential pressure across the systemic and pulmonary circulation were set to $+37 \pm 1$ °C, 6.2 ± 0.25 L/min, 78 and 33 ± 3 mmHg, respectively. The air pressure and volume in the compliance chamber, along with the tightness of the resistance clamp, were adjusted to achieve the target pressure head across both circulations. A static (“no pump”) control bag of the same blood pool was placed in a $+37 \pm 1$ °C water bath for each test. Pressure targets were based on the clinical SynCardia values¹⁴.

Sample and data collection

Data acquisition (DAQ) measuring boxes (NI-9237 and NI-9212) and DAQ-chassis (NI cDAQ-915; National Instruments Germany GmbH, Munich, Germany) collected flow, pressure, and temperature data.

Ports were cleared by discarding 0.5 mL of blood prior to blood sample collection. Hematology and pffHb were analyzed at all time points from a 1.5 mL sample. At 5, 180, and 360 min, an additional 2 mL sample was collected to analyze blood gas and VWF, Table 1.

Hemolysis Harboe assay

Plasma free hemoglobin was quantified using a spectrophotometric method based on the Harboe assay. Samples were processed by sequential centrifugation and stored at -80 °C prior to analysis. Hemolysis was expressed as the normalized index of hemolysis (mgNIH) as per ASTM F1841, accounting for hematocrit, flow rate, circuit

	Sampling Schedule (min)							
	Blood Pool	5	60	120	180	240	300	360
Total Hemoglobin (g/L)	x	x	x	x	x	x	x	x
Hematocrit (%)	x	x	x	x	x	x	x	x
pfHb (mg/dL)	x	x	x	x	x	x	x	x
Erythrocyte count ($10^{12}/L$)	x	x	x	x	x	x	x	x
Leukocyte count ($10^9/L$)	x	x	x	x	x	x	x	x
Platelet count ($10^9/L$)	x	x	x	x	x	x	x	x
Granulocyte count ($10^9/L$)	x	x	x	x	x	x	x	x
Monocyte count ($10^9/L$)	x	x	x	x	x	x	x	x
Lymphocyte count ($10^9/L$)	x	x	x	x	x	x	x	x
VWF		x			x			x
pH	x	x			x			x
Glucose (mmol/L)	x	x			x			x
Flow rate, Pressure head, Temp.		x	x	x	x	x	x	x

Table 1. Sampling schedule for all tests.

volume, and sampling interval. The modified index of hemolysis (MIH) was derived by normalization to baseline total hemoglobin⁸.

Acquired von Willebrand syndrome analysis

Blood samples were centrifuged at 3000 x g at +15°C for 15 min to collect the platelet-poor plasma supernatant, which was frozen within 1 h after withdrawal from test loop/static control and stored at -80°C until analysis. Thawed samples were subjected to ELISA analysis of VWF collagen-binding activity and antigen determination (Asserachrome 00239 VWF: CB, and Asserachrome 00942, respectively, Diagnostica Stago Asnières sur Seine, France). The amount of VWF was detected using bound anti-VWF conjugated to horseradish peroxidase that was catalyzed by its substrate TMB (3,3',5,5'-tetramethylbenzidine). Absorbance was measured using a SpectraMax 190 Microplate Reader operated with SoftMax Pro Software, version 7.0 (Molecular Devices, San Jose, CA, USA; <https://www.moleculardevices.com/products/microplate-readers/acquisition-and-analysis-software/softmax-pro-software>). Then the ratio of VWF: CB to VWF: Ag was determined. Blood subjected to continuous-flow left ventricular assist device (CF-LVAD) BPX-80 (Medtronic) pumped at a pressure head of 100 ± 3 mmHg with a target flow of 5 L/min was used as a positive control for VWF damage.

Statistical methods and analysis

For hemolysis and hematology, data are presented as mean $\pm$ standard deviation at each time point. Hematology parameters were expressed as a percentage of their respective blood pool baseline to evaluate relative pump-induced changes while minimizing inter-donor variability.

mgNIH and MIH data were compared using Mann-Whitney U tests. PfHb and hematology parameters over time were analyzed using linear mixed-effects models, with treatment group (SYN, RH and static), time (duration), and their interaction (time x treatment) included as fixed effects, and test (blood pool) included as a random effect to account for repeated measurements within donor pools. Type III analysis of variance with Satterthwaite's approximation was used to assess fixed effects. Where significant interactions were detected, pairwise comparisons between treatment groups at individual time points were performed using estimated marginal means with Tukey adjustment for multiple comparisons.

VWF function was evaluated at 360 min. The VWF: CB/VWF: Ag ratio was compared against the clinically relevant threshold of 0.8¹⁵ using one-sample t-tests. SYN and RH were compared using paired tests due to shared blood pools ($n=5$). Comparisons between pumps and the positive control (BPX-80), which were conducted subsequently using independent blood pools ($n=4$), were performed using Welch's t-tests. To account for donor variability between experimental sets, VWF ratios were normalized to their respective static controls prior to statistical analysis.

Results

Pump and rig parameters

SYN was operated at 125 bpm and 50% systole with passive adjustment of the stroke volume to achieve the highest possible flow at clinically relevant pressures for the systemic and pulmonary circulation. To match the same flow and pressure heads, RH was operated at 125 bpm, 54% systole, and full device stroke length (left pump: 26 mm, right pump: 22 mm).

SYN ($n=6$) had a mean flow of 6.27 ± 0.03 L/min (approximate stroke volume of 50 mL), systemic circulation pressure head of 77.4 ± 0.4 mmHg, pulmonary circulation pressure head of 33.2 ± 0.2 mmHg and 35.3 ± 1.4 °C, respectively. Corresponding RH values ($n=6$) were 6.28 ± 0.06 L/min, 78.0 ± 0.5 mmHg, 32.9 ± 0.4 mmHg, and temperature of 35.7 ± 1.3 °C, more detail shown in Table 2.

		Sampling Schedule (min)						
		5	60	120	180	240	300	360
Flow ($n=6$) L/min	SYN	6.39 ± 0.18	6.30 ± 0.31	6.29 ± 0.32	6.29 ± 0.36	6.23 ± 0.32	6.23 ± 0.31	6.22 ± 0.30
	RH	6.31 ± 0.22	6.25 ± 0.24	6.24 ± 0.22	6.24 ± 0.20	6.28 ± 0.19	6.29 ± 0.19	6.29 ± 0.19
Preload left ($n=6$) mmHg	SYN	7.48 ± 0.67	7.79 ± 0.82	7.54 ± 0.54	7.36 ± 0.25	7.05 ± 0.24	6.95 ± 0.38	7.15 ± 0.44
	RH	6.56 ± 1.17	7.28 ± 0.79	7.16 ± 0.53	6.86 ± 0.73	6.81 ± 0.45	6.64 ± 0.42	6.94 ± 0.42
Afterload left ($n=6$) mmHg	SYN	84.98 ± 0.33	84.55 ± 1.54	84.79 ± 0.82	84.65 ± 1.16	84.79 ± 0.95	84.71 ± 0.79	84.84 ± 1.47
	RH	84.33 ± 1.32	84.60 ± 1.08	84.96 ± 0.95	85.39 ± 0.90	85.43 ± 0.40	84.98 ± 1.03	84.66 ± 1.43
Pressure head left ($n=6$) mmHg	SYN	77.5 ± 0.9	76.8 ± 1.2	77.2 ± 0.5	77.3 ± 1.2	77.8 ± 0.8	77.8 ± 0.8	77.7 ± 1.3
	RH	77.8 ± 1.1	77.3 ± 0.8	77.8 ± 1.3	78.5 ± 0.9	78.6 ± 0.7	78.3 ± 0.7	77.7 ± 1.7
Preload right ($n=6$) mmHg	SYN	7.44 ± 1.41	7.20 ± 1.29	6.88 ± 0.97	6.87 ± 0.73	7.02 ± 1.06	6.95 ± 1.07	6.95 ± 1.21
	RH	6.66 ± 1.41	6.82 ± 1.13	6.76 ± 0.48	6.81 ± 0.51	6.50 ± 0.66	6.33 ± 0.85	6.18 ± 0.76
Afterload right ($n=6$) mmHg	SYN	39.84 ± 2.22	39.65 ± 0.91	39.96 ± 0.78	40.18 ± 0.94	40.21 ± 0.63	39.68 ± 1.04	40.00 ± 0.72
	RH	39.88 ± 1.41	40.06 ± 1.14	40.40 ± 0.80	39.79 ± 0.45	39.98 ± 1.22	39.22 ± 1.37	39.23 ± 1.45
Pressure head right ($n=6$) mmHg	SYN	32.4 ± 3.5	32.5 ± 1.1	33.1 ± 0.4	33.3 ± 1.5	33.2 ± 1.5	32.8 ± 2.0	33.0 ± 1.6
	RH	33.2 ± 1.9	33.2 ± 1.1	33.6 ± 1.0	33.0 ± 0.5	33.5 ± 1.2	32.9 ± 1.8	33.1 ± 1.9
Temp. ($n=6$) °C	SYN	33.1 ± 0.8	35.1 ± 0.2	36.2 ± 0.5	36.4 ± 0.4	36.4 ± 0.4	36.4 ± 0.4	36.4 ± 1.0
	RH	32.3 ± 1.6	35.0 ± 0.2	35.5 ± 0.5	36.0 ± 0.6	36.1 ± 0.4	36.1 ± 0.4	36.1 ± 0.4

Table 2. Mean flow, preload, afterload, pressure head and temperature for the SYN and RH. Data are mean ± SD ($n=6$).

Realheart TAH V11.4 prototype hemolysis significantly lower than SynCardia

The mean pfHb concentration ($n=6$) for the baseline was about 9.19 mg/dL at the start of the experiment, below the threshold of 20 mg/dL thus indicating good blood quality. The linear mixed-effects model demonstrated significant effects of time ($F(6,100)=125.79, p<0.001$), treatment group (SYN, RH, and static, $F(2,100)=1709.05, p<0.001$), and a significant time x treatment interaction ($F(12,100)=80.83, p<0.001$) on pfHb. These findings show that hemolysis increased over time and differed among SYN, RH and static, with treatment-related differences varying across test duration (Fig. 2A).

Pairwise comparisons revealed that SYN produced significantly higher hemolysis than RH and static at all time points after 5 min ($p<0.001$, Fig. 2A), also evidenced by the increased redness of plasma (Fig. 2B).

RH showed lower hemolysis in each experiment. The mean mgNIH ($n=6$) of SYN and RH were 226.0 ± 30.6 mg/100 L and 35.5 ± 4.3 mg/100 L, respectively ($p=0.002$; Fig. 2C). The ratio in mgNIH of SYN to RH was 6.36. The MIH calculated was 20.80 ± 2.65 for SYN and 3.27 ± 0.37 for RH ($p=0.002$; Fig. 2D).

SynCardia significantly increases apparent platelet counts compared to Realheart TAH V11.4 and static control

No significant effect of time ($p=0.116$) or interaction ($p=0.487$) was observed for erythrocytes (Fig. 3A), although a main effect of treatment group was detected ($p<0.001$).

Leukocyte (Fig. 3B) levels demonstrated significant effects of time ($p<0.001$), treatment ($p<0.001$), and a significant interaction ($p=0.0049$). Leukocyte counts for both devices were lower than static at 360 min.

Platelet (Fig. 3C) counts and mean platelet volume (MPV; Fig. 3G) were significantly affected by time ($p<0.001$), treatment ($p<0.001$), and their interaction ($p<0.001$). SYN exhibited significantly higher platelet counts and MPV values compared to RH and static from 60 min onward (60–360 min). This phenomenon has previously been attributed to the degradation of larger blood cells, resulting in cellular fragments being erroneously classified as platelets by the hematology analyzer¹⁶. The concurrent increase in MPV suggests alterations in platelet morphology and size distribution during circulation.

Granulocyte (Fig. 3D) counts demonstrated significant time, treatment, and interaction effects (all $p<0.001$). Both devices had lower granulocyte counts than static at 360 min.

Monocyte (Fig. 3E) levels were significantly influenced by time ($p=0.012$), treatment ($p<0.001$), and their interaction ($p<0.001$). Both devices had higher monocyte counts than static at 360 min. This granulocyte decrease and monocyte increase can be explained by a gradual leftward shift of the corresponding cellular peaks in the leukocyte histogram (Fig. 4A) of devices compared to static control.

Lymphocyte (Fig. 3F) levels demonstrated significant effects of time ($p=0.0147$), treatment ($p=0.0108$), and interaction ($p=0.0035$). No significant difference between devices and static control was observed.

Red cell distribution width (RDW; Fig. 3H) demonstrated significant effects of time ($p<0.001$) and treatment ($p<0.001$), whereas no significant interaction was observed ($p=0.8009$).

Acquired von willebrand syndrome analysis

At 360 min, the VWF: CB/VWF: Ag ratio remained significantly above the clinical threshold of 0.8¹⁵ for both SYN (1.01 ± 0.07 ; $p=0.002$) and RH (1.01 ± 0.06 ; $p=0.001$). No difference was observed between SYN and RH.

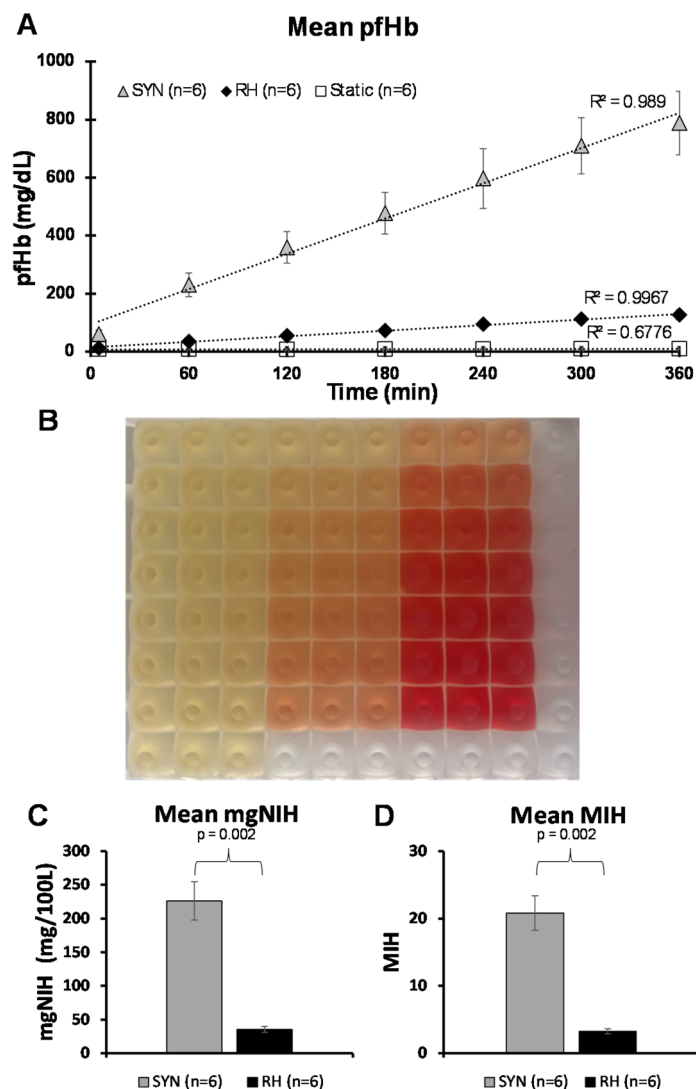


Fig. 2. Mean plasma free hemoglobin (pfHb) and hemolysis indices. (A) Mean pfHb for RH, SYN and static control. Results expressed as mean $\pm$ SD with the coefficient of determination ($n = 6$). (B) Representative image from one of the tests. Samples in triplicate (columns) prior to quantitative assessment. Left to right: Static, RH, SYN. From top to bottom: Each timepoint corresponds from 5 to 360 min. Last row corresponds to baseline control. (C) Mean mgNIH and (D) mean MIH calculated based on the ASTM F1841-97 (2017) formula averaged over the test. Results expressed as mean $\pm$ SD ($n = 6$).

In contrast, BPX-80 demonstrated lower ratios (0.78 ± 0.15). When normalized to respective static controls to account for blood pool variability, both SYN and RH showed significantly higher ratios compared to BPX-80 (null = 0.023 for both; Fig. 4B).

These findings indicate preserved VWF functional activity in both pulsatile devices compared to the positive CF-LVAD control.

Discussion

This is the first study to assess in vitro hemolysis of the complete SynCardia device under combined systemic and pulmonary conditions and compare it with the Realheart TAH. As the only FDA-approved TAH, SynCardia is the relevant clinical comparator.

The SynCardia hemolysis was 6 times greater than that of the Realheart TAH, which may be related to differences in valve design and movement. Published CFD studies on Realheart and SynCardia have demonstrated that the highest stresses occur in the vicinity of the valves and associated regurgitant flow through closed valves, with mitral valve leakage during systole and aortic valve leakage during diastole. In SynCardia, leakage occurs through the valve's central hinge hole and in the gap between the disc and the ring¹⁷, whereas in Realheart leakage occurs through the side and peripheral gaps of the bileaflet valves and has been identified as the principal source of hemolysis generation¹¹.

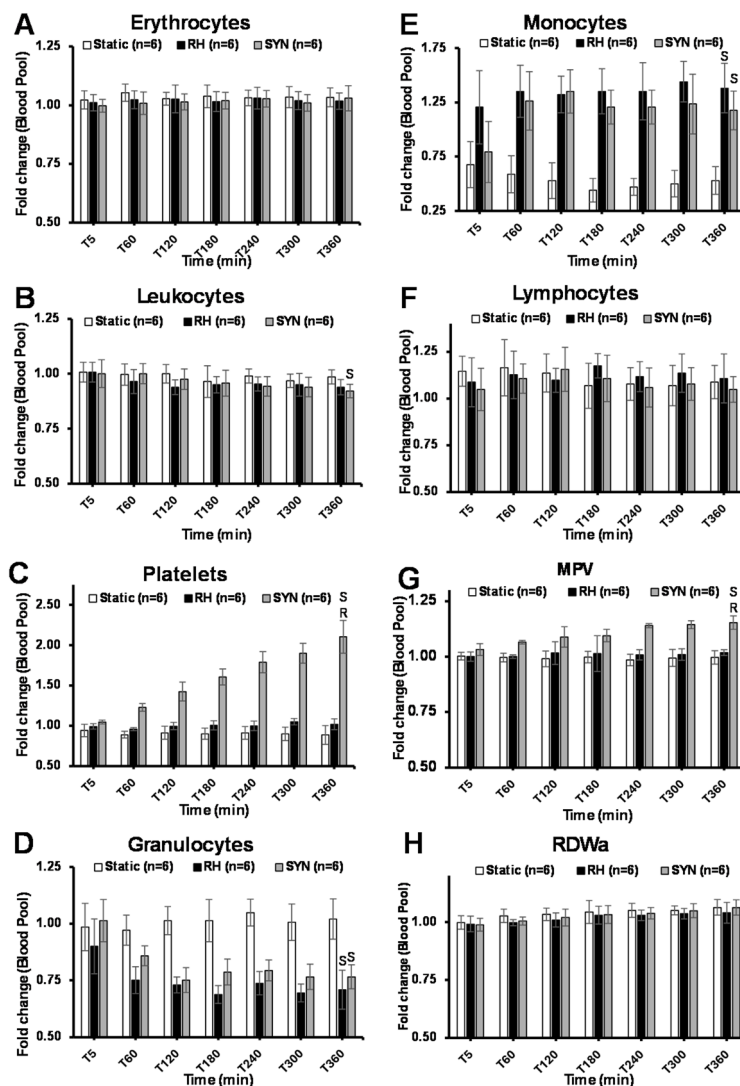


Fig. 3. Mean blood cell counts for the TAHs and static control. The fold change was derived from dividing each time point value by the blood pool baseline. Mean (A) erythrocytes, (B) leukocytes, (C) platelets, (D) granulocytes, (E) monocytes, (F) lymphocytes, (G) MPV, and (H) RDW for the static, RH and SYN. Results expressed as mean $\pm$ SD ($n=6$). S, significantly different from the static control; R, significantly different from the RH.

Valve design may contribute to the observed difference in hemolysis between the two TAHs. The single tilting disc MedHall valve released in 1977 was selected by SynCardia due to its mechanical resilience and has a maximum opening angle of 70–75°¹⁸. Blood must deflect around the disc, creating an asymmetric split flow pattern associated with turbulence and localized low-pressure regions prone to cavitation. In contrast, bileaflet valves create three flow channels, one central and two lateral¹⁹. In a comparison of the MedHall valve to earlier generation bileaflet valves, the closing velocity was 2–4.2 m/s versus 1.5–2.6 m/s, respectively. The MedHall valve also had longer cavitation duration compared to bileaflet valves²⁰. The Realheart TAH incorporates On-X bileaflet valves introduced in 1996, which open to 90 degrees and have been reported to generate less turbulence than earlier-generation bileaflet valves²¹.

Beyond valve design, the pumping principles also differ fundamentally. Unlike a diaphragm-driven pump, atrioventricular plane displacement generates flow through translation of the valve plane itself, producing flow patterns that more closely resemble the native heart. Previous CFD and 4D-flow MRI studies of the Realheart TAH demonstrated physiologic vortex formation, efficient ventricular washout (87% after four cycles), no evidence of stagnant regions, and low stasis comparable to native hearts^{13,22}. In addition, the motor-driven atrioventricular plane provides the ability to prescribe motion profiles throughout the cardiac cycle. Since low closing velocity and gentle deceleration are important for reducing valve cavitation²³, the ability to program atrioventricular plane motion may offer an additional advantage, although the impact of this feature on valve closing dynamics has not yet been quantified.

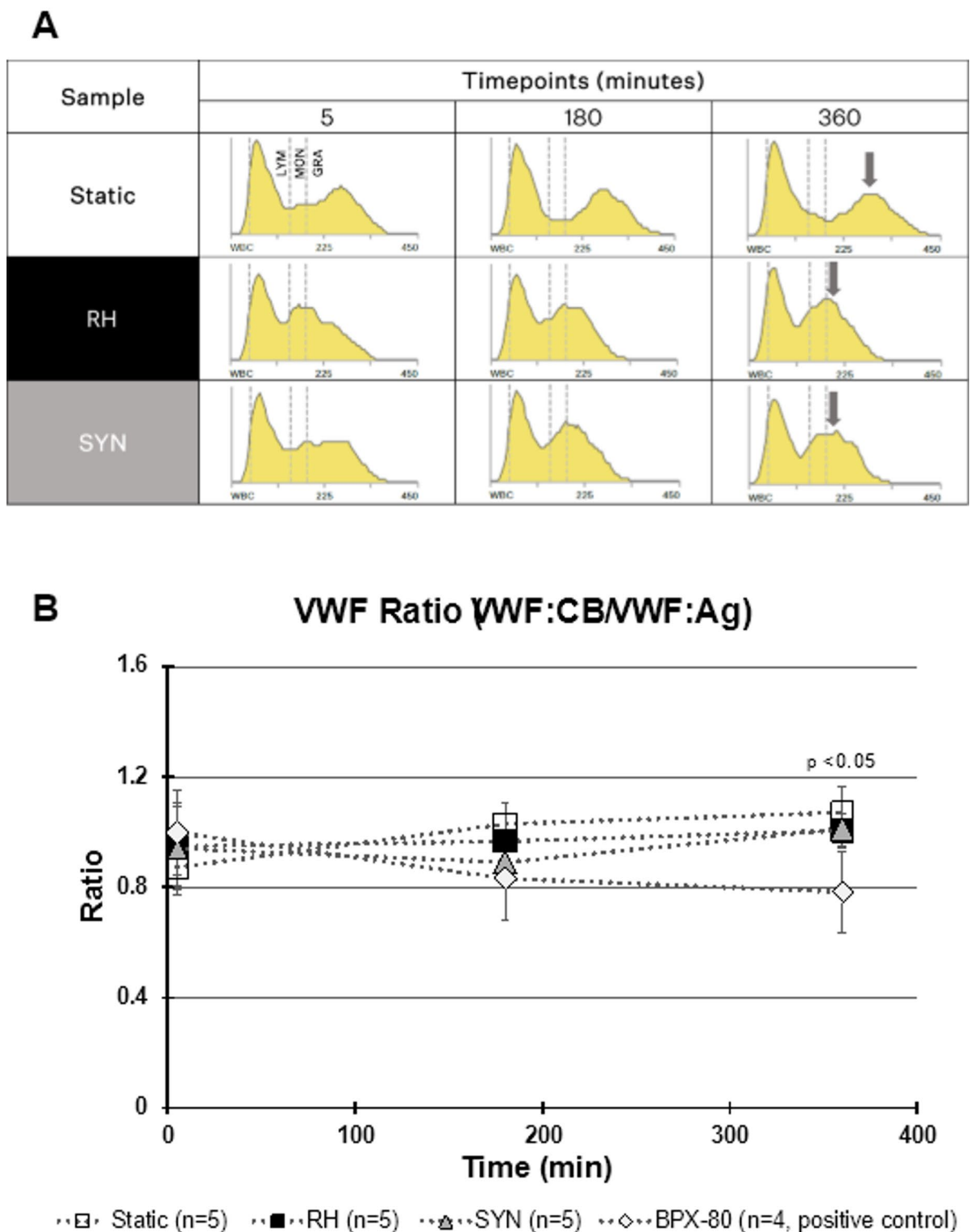


Fig. 4. Leukocyte histograms and VWF ratio (VWF: CB/VWF: Ag). (A) Leukocyte histograms for 5-, 180- and 360-min samples for static control, RH and SYN. The x-axis depicts the cell size in femtoliters (fL) and the y-axis depicts the cell count. The dotted lines are discriminator bars and separate cell types by size with granulocytes being the largest and lymphocytes being the smallest cells. The first peak corresponds to lymphocytes, and the second to granulocytes. Compared with the static control, the TAH samples exhibit a gradual leftward shift of the granulocyte peak entering the monocyte and lymphocyte gates. Arrows in the 360 min sample indicate this shift. (B) VWF ratio (VWF: CB/VWF: Ag) of static control, RH and SYN. Results expressed as mean $\pm$ SD ($n = 5$). A BPX-80 continuous-flow pump was subsequently used as positive control. Results expressed as mean $\pm$ SD ($n = 4$).

Although hemolysis in SynCardia recipients is not commonly listed among top adverse events in the MAUDE database, clinical reports show significant SynCardia hemolysis^{3,6} and also association of free hemoglobin exposure with downstream organ effects, including renal hemosiderosis²⁴. The plasma free hemoglobin observed in an in vitro experiment is generally greater than that observed in vivo where the hemoglobin is scavenged and metabolized and therefore needs to be evaluated alongside additional parameters. Thus, the average pfHb observed in SynCardia patients has only been around 7–20 mg/dL, but the hematocrit decreased at 1 month post-operatively versus baseline (~22 vs. ~35%)^{3,25}, and LDH has been markedly elevated at ~1,100 U/L³, indicating red cell destruction. Hemolysis, in combination with hypoperfusion (which can be the case in SynCardia patients who suffer anemia and mesenteric ischemia²⁶ may be a driver towards multiorgan failure – the most common cause of death⁵. However, TAH recipients are often more critically ill than LVAD recipients.

The hematology findings are consistent with previous mechanical circulatory support studies⁸. Reduced leukocyte counts and increases in platelet counts and MPV are consistent with cellular fragmentation and analyzer misclassification rather than definitive proof of this mechanism. This interpretation is supported by previous studies of mechanical circulatory support devices that used regular and imaging flow cytometry in combination with scanning electron microscopy to demonstrate that leukocyte fragments may be counted by automated hematology analyzers²⁷. Interestingly, the leukocyte populations most affected differed between studies. Whereas Chan et al. reported the greatest reduction in lymphocyte counts, Radley et al.¹⁶ subsequently observed that granulocyte loss was the predominant finding in blood exposed to the HVAD and HeartMate II, consistent with the predominance of granulocyte-associated changes observed in the present study. These observations suggest that the specific leukocyte populations affected may depend on device-related factors and flow conditions rather than reflecting a universal pattern of leukocyte injury. Although their clinical significance is uncertain, they may correspond to increased circulating microparticles or extracellular vesicles in vivo. LVAD-derived microparticles (MPs) have been linked to impaired flow-mediated vasodilation²⁸ and thrombosis²⁹.

VWF was preserved in both pulsatile TAHs, with no reduction in the VWF: CB/VWF: Ag ratio relative to static control, thus maintaining VWF functionality. In contrast, the continuous-flow BPX-80 showed reduced ratios, consistent with previous findings that pulsatile flow limits VWF degradation. That pulsatility is protective of VWF has been demonstrated in LVAD patients with the pulsatile HeartMate XVE in comparison to the continuous flow HeartMate II³⁰ and in a comparison between HeartMate II and SynCardia patients³¹. Pulsatile flow has been proposed to protect VWF through two mechanisms: (1) prevents VWF unfolding and thus exposure of ADAMTS-13 cleavage sites³², and (2) by not providing the mechanical stimulation of downstream endothelium to release more VWF into the circulation³³.

The modified mock circulation loop enabled evaluation of complete TAH devices under full-body circulation conditions. In this study, SynCardia was operated within its typical clinical range (120–130 bpm). Although the device is advertised to deliver flows up to 9.5 L/min, this was not achievable under the physiological preload conditions used here. At normal central venous pressure (CVP), the maximal attainable flow was 6.2 L/min, corresponding to an optimal filling volume of 50–55 mL per ventricle. Higher flows approaching 9.5 L/min require non-physiological conditions, including heart rates near 150 bpm and near-maximal ventricular filling (63 mL) supported by elevated preload (CVP ~20 mmHg). Such settings are undesirable clinically, as they represent an unphysiological loading state. Therefore, for the purposes of hemocompatibility testing in accordance with ASTM, the highest physiologically achievable flow – 6.2 L/min – was selected and matched between devices.

Both devices maintained appropriate differential pressures, although the SynCardia loop required longer to reach target temperature due to longer tubing lengths. Because elevated temperature increases hemolysis, this delay favoured SynCardia and therefore does not undermine the observed differences.

In summary, the modified test system enabled comprehensive hemolysis evaluation of full TAH devices under combined systemic and pulmonary loading. The Realheart TAH prototype V11.4 generated markedly lower hemolysis and reduced cellular debris formation compared to SynCardia, while maintaining VWF activity. These findings suggest potential clinical advantages, including reduced risk of hemolysis-associated organ injury, while preserving the protective pulsatile effects on VWF.

Limitations

This study has several limitations. First, the in vitro mock circulation model, although designed to approximate physiological systemic and pulmonary conditions, does not capture the full complexity of the in vivo environment, including endothelial and organ-level responses. Second, pooled healthy donor blood under controlled laboratory conditions does not fully reproduce the complex inflammatory, coagulopathic, anemic and endothelial alterations present in patients with end-stage heart failure requiring TAH support, and the 6-hour duration captures acute hemocompatibility effects. Inflammation has been reported to reduce red blood cell deformability³⁴ and may increase susceptibility to mechanically induced hemolysis³⁵, while dysfunction of the endothelium, which is the main producer of circulating VWF, may alter von Willebrand factor levels³⁶. Third, despite efforts to standardize flow and pressure conditions, minor differences in circuit configuration were present. However, identical tubing materials commonly used in cardiopulmonary bypass circuits were applied, and slightly lower operating temperatures in the SynCardia loop (conditions generally associated with reduced hemolysis) would be expected to bias results in favor of SynCardia. Finally, the sample size reflects standard ASTM-based paired hemocompatibility testing and supports controlled comparative evaluation.

Data availability

The datasets analyzed in the current study are available from the corresponding author upon reasonable request.

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Author contributions

S.F.Z. and I.L.P. conceived the study. S.F.Z., K.J.V., T.F., O.V., and I.L.P. contributed to the methodology. S.F.Z., K.J.V., T.F., O.V., and I.L.P. performed the investigation. S.F.Z. and K.J.V. curated the data. S.F.Z., K.J.V., and I.L.P.

performed the formal analysis. T.F., O.V., and I.L.P. supervised the study. I.L.P. acquired funding and managed the project. S.F.Z. and I.L.P. wrote the original draft. All authors reviewed and edited the manuscript and approved the final version.

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Declarations

Competing interests

The authors declare no competing interests.

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